Bill Vo

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Education

University of British Columbia Sep. 2023 – Present

Bachelor of Science in Computer Science (Honour)

- Coursework: Data Structure and Algorithm, Data Analysis, Data Processing, Software Engineering, Machine Learning / Deep Learning / AI, Parallel Computing

Experience

Software Engineer Intern Jan. 2026 – Present

Safari AI *Vancouver, BC, Canada*

- Proposed and built an internal automation for Safari's **AI-automation (GTM) initiative** called **Pipeline Tracker** (**React**, **FastAPI**, **HubSpot CRM API**, **S3**) that replaced a manual sales-pipeline reporting process with a **change-data-capture** layer producing analyst-ready data.
- Delivered ahead of schedule an **executive-summary** feature I proposed beyond assigned scope, adopted by customers who had not requested it.
- Utilized **AWS Lambda** behind **API Gateway** (instead of an always-on **EC2** instance) to serve NFC-sticker redirects engine which saves a lot of money including compute cost and engineering cost (which is needed to maintain EC2)
- Chose **FastAPI** with the lightweight **Strands Agents** SDK (good for per-conversation **session persistence** in PostgreSQL) over a full agent framework like **LangChain** for the platform's LLM features: our flows are fixed sequences, not open-ended tool exploration, so deterministic **Python** routes the workflow while the model only narrates results.

VLM Research Assistant Sep. 2025 – April 2026

UBC Okanagan *Vancouver, BC*

- Conducting research on **fine-tuning and evaluation of Remote Sensing VLMs** such as **RS-LLaVA**, **RSGPT**, and **RemoteCLIP** on datasets like **RSICD**, **UCM Captions**, and **RSVQA**.
- Evaluate the output score on these metrics: **BLEU-1/2/3/4, METEOR, ROUGE-L, CIDEr**

Software Developer Intern Sep. 2025 – December 2025

UBC Okanagan *Vancouver, BC*

- Developed an **MCP tool** (using Python, FastAPI) that translates **natural language queries into Elasticsearch DSL**, integrating open-source models such as **Qwen-2.5** and **TinyLlama** for keyword extraction and query generation. This helped researchers in STASER Lab easily access data just by using NLQ.
- Designed and tested **modular pipelines** connecting the MCP client with Elasticsearch indices and implemented evaluation scripts for semantic correctness and retrieval accuracy.

Projects

ScholarDash — Research Funding Intelligence Platform | ***Python, DuckDB, dbt, Airflow, Streamlit*** 2026

- Built an end-to-end data pipeline ingesting **1,000+ federal research grants** from **NSF** and **NIH APIs** into a **DuckDB** warehouse with dual persistence (database + JSON archive), rate limiting, and per-record quality scoring.
- Designed a **star schema** with **14 dbt models** (2 staging, 5 dimensions, 1 fact, 5 marts) and **25+ automated tests** covering uniqueness, referential integrity, and value ranges.
- Orchestrated the pipeline with **Apache Airflow** via **Docker Compose** (4 services), running an **8-task daily DAG** with retry logic and DuckDB single-writer concurrency management.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, HTML/CSS, Bash, C/C++
Frameworks & Libraries: FastAPI, Flask, React, Next.js, Node.js/Express, PyTorch, LangChain, Pydantic, Pandas, NumPy, Streamlit, Plotly
ML/AI: LLM APIs (GPT-4o), RAG, LoRA, Embeddings (bge-small-en), ONNX Runtime, Evaluation (BLEU, ROUGE, CIDEr, METEOR, SPICE)
Platforms & Tools: Apache Airflow, dbt, DuckDB, HubSpot CRM, Docker, CI/CD, Git, Kubernetes (CronJob), Postman, MongoDB, SQLite, Databricks, GCP, AWS
Certification: ETL in Python and SQL (LinkedIn), Databricks Fundamentals, SQL Intermediate (HackerRank)